

## Detection of active-site peptides of kinases

\*use HPLC water for all steps

**\*when measuring inhibitor dry stocks for DMSO solutions, do not try to measure less than 2 mg of compound. The error from measuring small amounts of compound can drastically affect the final concentrations tested in your assay.**

### Labeled protein reduction and alkylation

1. Dilute light and heavy lysate with reaction buffer to 2 mg/mL and transfer 250  $\mu$ L (0.5 mg) to a **screw-cap** microcentrifuge tube (Sarstedt D-51588).
  - Reaction buffer: PBS + 20mM  $MgCl_2$  + Pierce Protease Inhibitor Tablet
2. **Optional:** Pre-treatment of lysate with compound. Add 50X compound stock to each sample (5  $\mu$ L), mix gently and incubate at room temperature for 30 minutes.
3. Add 5  $\mu$ L of 0.50 mM desthiobiotin ATP probe (10uM Final) to each sample. Incubate at room temperature for 30 minutes.
  - Probe is moisture sensitive. Remove only aliquot of probe that you are using from -80C freezer. Do not subject probe to multiple freeze-thaw cycles. Equilibrate to room temperature before opening and prepare immediately before use.
  - 0.50 mM = One aliquot (12.6 $\mu$ g) ATP probe diluted in 20 $\mu$ L ultrapure water

### MeOH/ $CHCl_3$ ppt to remove excess probe and desalt

1. Add 2 mL (4x volume) MeOH to an empty dram vial
2. Add 0.5 mL (1x volume)  $CHCl_3$
3. Add 1.5 mL (3x volume)  $H_2O$
4. Transfer matched light and heavy samples into a single 8mL/2dram glass vial with solvents
  - Probe and compound CAN cross-react with proteins from both light and heavy lysates if not quenched immediately in solvent. This will skew final peptide ratios.
5. Centrifuge 1,400 g x 3 min at 4°C and allow to settle for 3 min or until the top layer is clear.
6. Remove top and bottom layers, leaving protein interface
7. Transfer interface to **screw-top** microcentrifuge tube in 600  $\mu$ L MeOH
8. Add 150  $\mu$ L  $CHCl_3$ , vortex
9. Add 600  $\mu$ L  $H_2O$ , vortex, centrifuge 1,400 g x 3 min
10. Remove top and bottom layers, leaving protein interface
11. Add 600  $\mu$ L MeOH, sonicate with probe sonicator 3 pulses (output 3)
12. Centrifuge 14,000 rpm 5 min to pellet protein, remove MeOH

#### D. Reduction, Alkylation

1. Re-suspend pellet in 500  $\mu$ L of 6 M urea/25 mM ammonium bicarbonate (ambic)  
*3.6g Urea (MW=60.06) in 1 mL 250 mM  $\text{NH}_4\text{HCO}_3$  stock*  
*(MW  $\text{NH}_4\text{HCO}_3$ =79.06; 198 mg into 10 mL gives 250 mM stock, make fresh)*  
*Bring volume up to 10 mL with milliQ water*
2. Sonicate with probe sonicator 3 pulses (output 3)
3. Add 5  $\mu$ L 100x (1M, 15.4 mg/100  $\mu$ L in  $\text{H}_2\text{O}$ ) DTT, giving 10 mM final (**make fresh**)
4. Incubate 65°C 15 min, let cool to RT or cool briefly on ice
5. Add 40  $\mu$ L 50x (0.5 M, 28 mg/300  $\mu$ L in  $\text{H}_2\text{O}$ ) IAA, giving 40 mM IAA (**make fresh**)
6. Incubate RT 30 min, in the dark

#### MeOH/ $\text{CHCl}_3$ ppt to remove excess probe and desalt

1. Transfer samples to 8 mL/2 dram glass vial
2. Add 2 mL (4x volume) MeOH
3. Add 0.5 mL (1x volume)  $\text{CHCl}_3$ , vortex
4. Add 1.5 mL (3x volume)  $\text{H}_2\text{O}$ , vortex
5. Centrifuge 1,400 g x 3 min at 4°C and allow to settle for 3 min or until the top layer is clear.
6. Remove top and bottom layers, leaving protein interface
7. Transfer interface to **Sarstedt screw-cap** tube in 600  $\mu$ L MeOH
8. Add 150  $\mu$ L  $\text{CHCl}_3$ , vortex
9. Add 600  $\mu$ L  $\text{H}_2\text{O}$ , vortex, centrifuge 1,400 g x 3 min
10. Remove top and bottom layers, leaving protein interface
11. Add 600  $\mu$ L MeOH, sonicate with probe sonicator 3 pulses (output 3)
12. Centrifuge 14,000 rpm 5 min to pellet protein, remove MeOH

#### Trypsin digestion

1. Re-suspend protein pellet in 500  $\mu$ L 25 mM ambic.
2. Reconstitute 20  $\mu$ g of sequence-grade trypsin/lys-c (1 vial) with 40  $\mu$ L of 25 mM ambic.
3. Add 7.5  $\mu$ g (15  $\mu$ L) Trypsin/Lys-c to sample and incubate at 37 C for 3 hr.

#### Labeled protein enrichment and elution

1. Remove 100  $\mu$ L aliquot of 50% avidin slurry per sample and place in 15 mL conical. Wash 3X by adding 10 mL of PBS, centrifuging at 1,400 g for 1 min., and decanting.
2. Resuspend washed beads in 1 bed volume of PBS and transfer equivalent amount of beads to all samples in 15 mL conicals. Add PBS for a total volume of 5.5 mL per sample.
3. Incubate RT, rotating for 1 hr
4. Spin 1,400 g for 1 min to pellet, remove ~90% of supernatant by decanting **ONCE**  
\*\*\*Do not decant more than one time!!! Your beads will become re-suspended, and you will dump out all of the beads on the second pour!

### Wash beads/elution

1. Wash 3X with 25 mM ambic by adding 10 mL, decanting **ONCE**, and centrifuging 1,400 g x 3 min.
2. Wash 3X with LC-MS water by adding 10 mL, decanting **ONCE**, and centrifuging 1,400 g x 3 min.
3. Transfer beads in 200  $\mu$ L of LC-MS grade water into **low-bind** microcentrifuge tubes.
4. Spin 1,400 g for 3 min to pellet, remove ~90% of supernatant (as much as you possibly can) by pipetting with gel tip.
5. Elute peptides by adding 100  $\mu$ L of elution buffer (50% ACN, 0.1% FA), incubating for 3 min., then spinning 1,400 g for 3 min. Wait 1 min. to let beads settle after centrifuge has stopped. Now transfer 75  $\mu$ L (of the original 100  $\mu$ L) of eluate to a new **low-bind** tube using a gel loading pipette tip. Be EXTREMELY careful. DO NOT TRANSFER BEADS.
6. Repeat the elution step (Step #5) 2X, instead starting with 75  $\mu$ L of elution buffer on the 2nd and 3rd elution. Collect all eluent into a single tube and spin down quickly on the bench-top centrifuge (~225  $\mu$ L total).
7. Spike in synthetic peptide standards for normalization and QC purposes: prepare a 100 fmol/ $\mu$ L Angiotensin/Vasoactive (AV) standard solution and add 2.5  $\mu$ L of peptide mix into each sample. \*after re-suspending samples in 50  $\mu$ L of mobile phase in step 10, the AV concentration in samples is 5 fmol/ $\mu$ L and injection of 2  $\mu$ L onto the LC-MS results in analysis of 10 fmol of AV peptides.
8. Poke holes from inside towards outside of microcentrifuge cap (minimize plastic debris in sample).
9. **Snap freeze peptides in liquid N<sub>2</sub>** (speeds up the drying down process) and dry down on speedvac. **Do not allow liquid N<sub>2</sub> to enter into tube.**  
**\*AVOID performing the speedvac drying step overnight because of peptide adsorption issues described in step 11 below**
10. Re-suspend peptides in 50  $\mu$ L of mobile phase A (LC-MS water + 0.1% FA) and analyze 2  $\mu$ L of sample on LC-MS.  
**\*note that 2  $\mu$ L injection volume is generally recommended; the volume may need to be adjusted based on proteome input and protein/peptide coverage**
11. **DO NOT store peptides in low-bind tubes because peptides will adsorb to the plastic and reduce signals on the LC-MS. After reconstituting peptides with mobile phase A, transfer 25  $\mu$ L aliquots of peptides into 2X amber vials with spring-loaded glass inserts for storage. One vial is stored as backup while the other vial will allow 2X injections on the QE+ protein nanoLC-MS.**

## Stage Tip cleanup of peptides

### A. Protocol to prepare the StageTip

#### Materials:

Empore Extraction Disks:

Octadecyl (C18)-bonded silica, part no: 2215, Vendor: 3M, St. Paul, MN



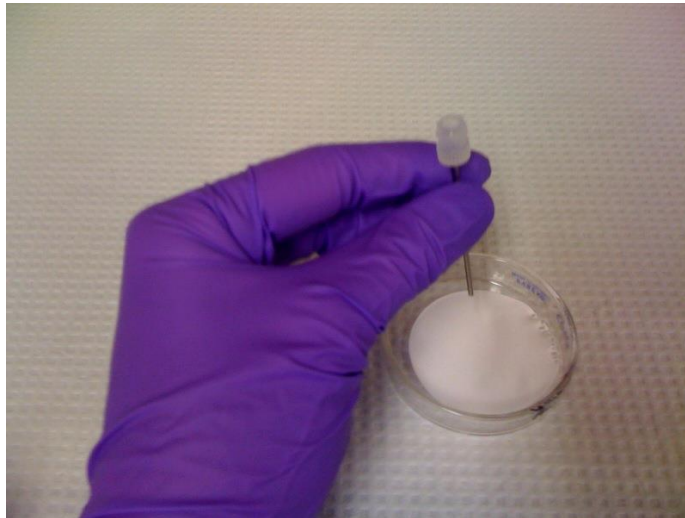
#### 200 µl Pipette tips

Needle: Kel-F Hub (KF), point style 3, gauge 16(1.19mm) part no: 90516

Plunger Assembly: N, RN, LT, LTN for model 1702 (25 µl)



1. Place one Empore extraction disk into a petri dish.
2. Use the Needle to pick up the stationary phase from the disk.
3. Use the plunger to insert into the 200 µl pipette tip.



### B. C18-StageTips for clean-up and concentration

**\* Use gel loading tips to load the solvents and the sample right on top of the disk! Do not dry the disk at any time! Disk without any liquid will dry out in 1-3 min! Repeat beginning from Step 1 if you dry the disk.**

1. Condition column by wetting disk with 20  $\mu$ L methanol (0.9G for 1 min)
    - a. Do not let tips dry completely
  2. Add 20  $\mu$ L Buffer B (80% ACN + 0.1% FA) to stage tip. Spin at 900 x *g* for 1 min.
  3. Add 20  $\mu$ L Buffer A (Water + 0.1% FA) to stage tip. Spin at 900 x *g* for 1 min.
  4. Load sample directly to the disc using gel loading tip. Spin at 900 x *g* for 1 min.
  5. Wash using 20  $\mu$ L Buffer A. Spin at 900 x *g* for 1 min.
 

\*Repeat step three times. This step might take longer than 1 minute, but check it periodically and ensure the stage tip does not dry.
  6. Switch to low-bind tube.
  7. Elute using 20  $\mu$ L buffer B. Spin at 900 x *g* for 1 min.
  8. \*Repeat step three times. This step might take longer than 1 minute, but check it periodically and ensure the stage tip does not dry.
  9. Spike in synthetic peptide standards for normalization and QC purposes: prepare a 100 fmol/ $\mu$ L Angiotensin/Vasoactive (AV) standard solution and add 2.5  $\mu$ L of peptide mix into each sample. \*after re-suspending samples in 50  $\mu$ L of mobile phase in step 12, the AV concentration in samples is 5 fmol/ $\mu$ L and injection of 2  $\mu$ L onto the LC-MS results in analysis of 10 fmol of AV peptides.
  10. Poke holes from inside towards outside of microcentrifuge cap (minimize plastic debris in sample).
  11. **Snap freeze peptides in liquid N<sub>2</sub>** (speeds up the drying down process) and dry down on speedvac. **Do not allow liquid N<sub>2</sub> to enter into tube.**
- \*AVOID performing the speedvac drying step overnight because of peptide adsorption issues described in step 13 below**
12. Re-suspend peptides in 50  $\mu$ L of mobile phase A (LC-MS water + 0.1% FA) and analyze 3  $\mu$ L of sample on LC-MS.
 

\*note that 3  $\mu$ L injection volume is generally recommended; the volume may need to be adjusted based on proteome input and protein/peptide coverage
  13. **DO NOT store peptides in low-bind tubes because peptides will adsorb to the plastic and reduce signals on the LC-MS. After reconstituting peptides with mobile phase A, transfer 25  $\mu$ L aliquots of peptides into 2X amber vials with spring-loaded glass inserts for storage. One vial is stored as backup while the other vial will allow 2X injections on the QE+ protein nanoLC-MS.**

**Revised: 08/01/2022**